بسم الله الرحمن الرحيم

Topic:

Aromaticity

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Aromaticity

- **Literal Meaning:** The literal meaning of "aromaticity" is **"fragrance,"** but the word has a special meaning in chemistry.
- There is no general relationship between aromaticity as a chemical property and the olfactory properties (relating to the sense of smell) of such compounds.

- **Definition:** "Aromaticity is a **property of conjugated cycloalkenes** in which the stabilization of the molecule is enhanced due to the ability of the electrons in the π orbitals to delocalize." This act as a framework to create a planar molecule.

OR

- **"In organic chemistry,** the term aromaticity is used to describe a **cyclic** (ring-shaped), **planar** (flat) molecule with a **ring of resonance bonds** that exhibits **more stability** than other geometric or connective arrangements with the same set of atoms."
- **As a result of their stability,** it is very difficult to cause aromatic molecules to break apart and to react with other substances.

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- **Aryl group:** "A functional group or other substituent that is aromatic is called

Characteristics of aromatic (aryl) compounds:

- A short list of rules, discovered by Eric Huckel in the 1930's, lists the properties of aromatic compounds. In order to be aromatic, a molecule must possess the following four structural characteristics:

1. Aromatic compounds are **cyclic** i.e. contributing atoms are arranged in one or more rings.

2. Aromatic compounds contain a **delocalized conjugated π-electron system,** most commonly in an arrangement of alternating single and double bonds. This conf guration allows for the electrons in the molecule's pi system to be delocalized around the ring, increasing the molecule's stability.

3. They have a **coplanar structure,** with all the contributing atoms in the same plane.

4. They contain a number of π delocalized electrons that is even, but not a multiple of 4. That is, **4n + 2 π-electrons,** where n = 0, 1, 2, 3, and so on. This is known as **Hückel's rule.**

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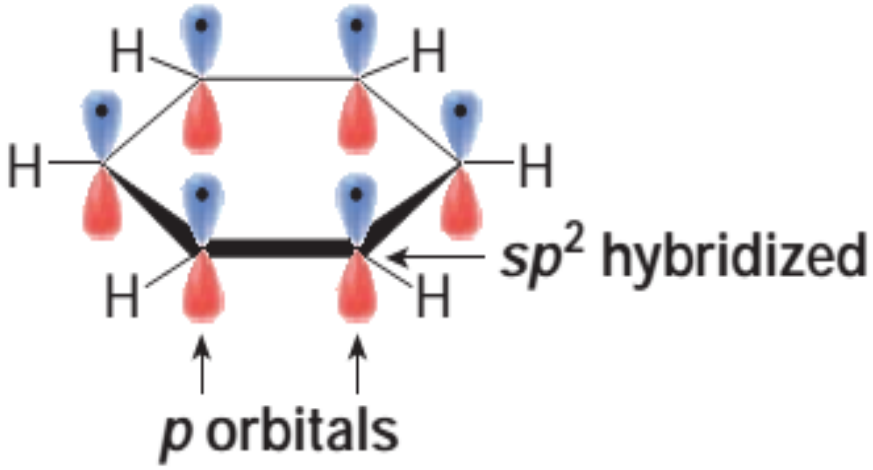
Example of an Aromatic Compound:

- **Benzene (C_6H_6)** is the **simplest aromatic hydrocarbon** (or arene) and one of the most commonly encountered aromatic systems of compounds in organic chemistry is based on **derivatives of benzene**.

- The word "aromatic" is occasionally used to refer informally to benzene derivatives.

- Nevertheless, many **non-benzene aromatic compounds also exist.** In living organisms, for example, the most common aromatic rings are the double-ringed bases in RNA and DNA.



Benzene—A planar molecule

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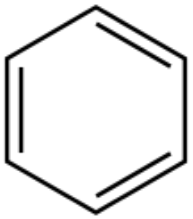
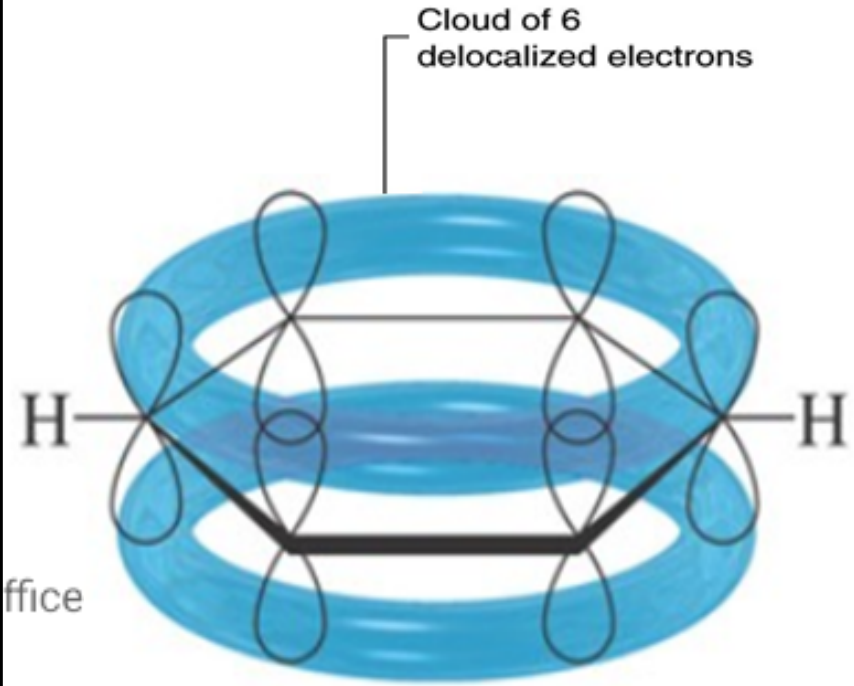


Benzene

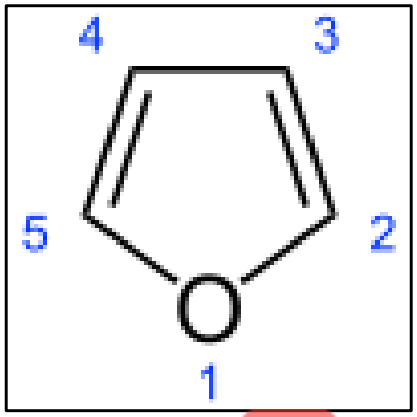


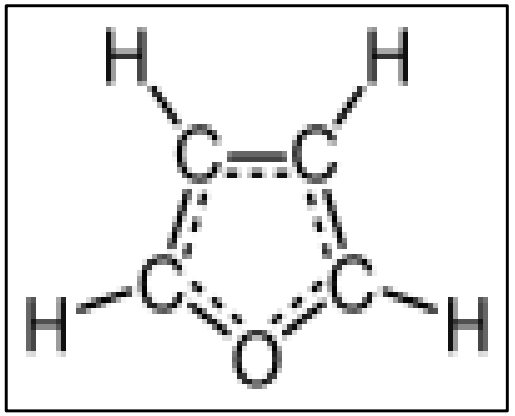
Example of a Heterocyclic (Non-Benzene) Aromatic Compound:

- In **furan,** the oxygen atom is sp^2 hybridized.

- One lone pair is in the π system (which participate in aromaticity) and the other in the plane of the ring (analogous to the C–H bond in the other positions).

- There are 6 π-electrons (2 electrons in the form of a lone pair and 4 electrons in two of the respective π - bonds) i.e., it follows 4n + 2 rule, so furan is aromatic.





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Furan

Anti-Aromatic & Non-Aromatic Compounds

- According to Huckel's rule, if a planar molecule has 4n + 2 π-electrons, it is **aromatic,** but if it has 4n π-electrons and has characteristics 1–3 above, the molecule is said to be **antiaromatic.**

- Antiaromatic compounds are, in general, destabilized and more reactive than expected.

Examples of Aromatic, Antiaromatic and Non-aromatic Compounds:

- **In benzene** each of the three double bonds contributes 2 pi-electrons over 6 atoms, for a total of 4 x 1 + 2 = 6 electrons, in a ring, in a pi-orbital that is planar. Therefore it is **aromatic.**

- **Cyclobutadiene** is **antiaromatic**, since the number of π delocalized electrons is 4, which of course is a multiple of 4. The cyclobutadienide (2) ion, however, is aromatic (6 electrons).

- Molecules that could be antiaromatic will tend to alter their electronic or conformational structure to avoid this situation, thereby becoming **non-aromatic.**

- For example, **cyclooctatetraene (COT)** distorts itself out of planarity, breaking π overlap between adjacent double bonds.

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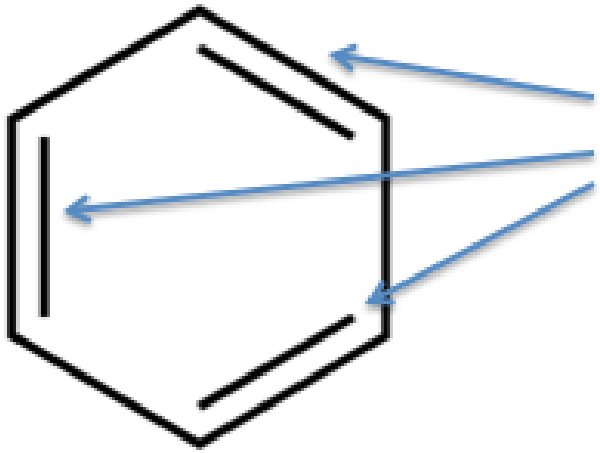
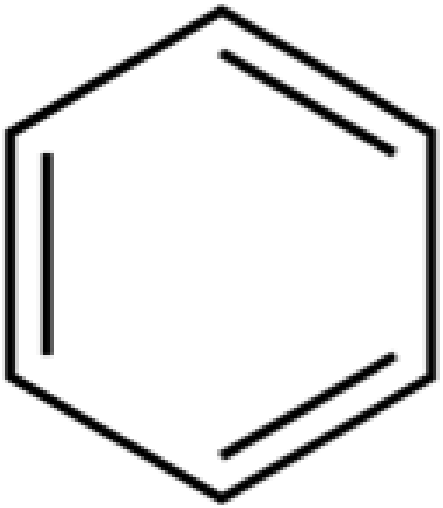
Benzene



3 double bonds = 6 pi electrons

Benzene



Since benzene has 6 pi electrons:

$4n + 2 = 6$

Find n:

$4n + 2 = 6$

$4n = 6-2$

$4n = 4$

$\mathbf{n = 1}$

An aromatic compound follows Huckel's rule if n is equal to zero or a positive whole number.

Benzene is aromatic

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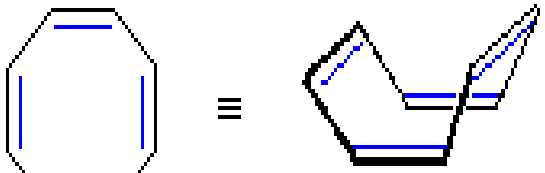
cyclobutadiene (Antiaromatic)	**1,3,5,7-Cyclooctatetraene (Non-aromatic)**

- Conjugated ring systems having 4n π-electrons (e.g. 4, 8, 12 etc. electrons) not only fail to show any aromatic properties, but appear to be less stable.
- As noted above, 1,3,5,7-cyclooctatetraene is non-planar and adopts a tub-shaped conformation.

- **Examples** of 8 and 12-π-electron systems are shown below, together with a similar 10 π-electron aromatic compound.

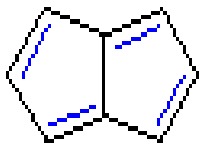


tub conformer (non-planar)

cyclooctatetraene stable b.p. 142 °C

gives alkene-like reactions

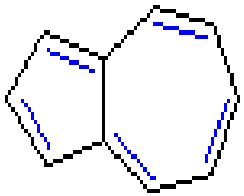
8 π-electrons



pentalene (planar)

very unstable

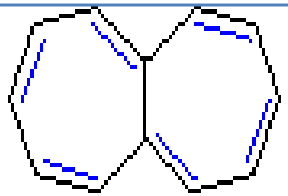
8 π-electrons



azulene (planar)

aromatic stability

10 π-electrons



heptalene (planar)

very unstable

12 π-electrons

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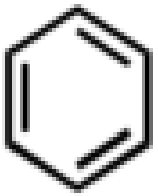
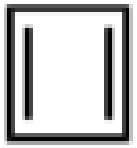
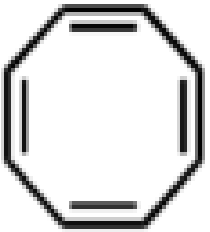
Aromatic	Anti-Aromatic	Non-Aromatic
unusually stable	***unusually unstable***	***everything else***
•Cyclic	•Cyclic	*Fails any one of the 4 criteria on the left*
•Conjugated	•Conjugated	
•(4n+2) Pi electrons	•(4n) Pi electrons	
•Flat	•Flat	
e.g. 	e.g. 	e.g. 
benzene	**cyclobutadiene**	**cyclooctatetraene**
(resonance energy of 36 kcal/mol)	(only stable below –100 °C)	(adopts a tub-like shape to "escape" anti-aromaticity)

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Types of aromatic compounds

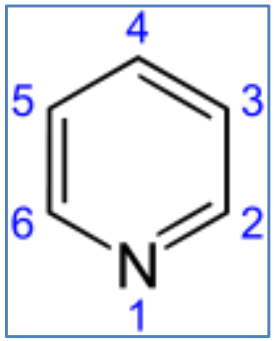
[1]. Simple Aromatic Compound Like Benzene.

[2]. Heterocyclics:

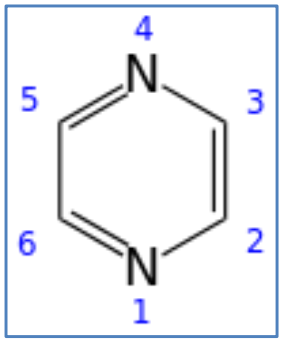
- In heterocyclic aromatic compounds, one or more of the atoms in the aromatic ring is of an element other than carbon.

- This can lessen the ring's aromaticity, and thus increase its reactivity.

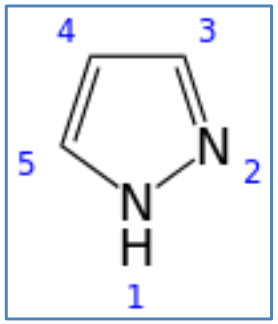
Examples:



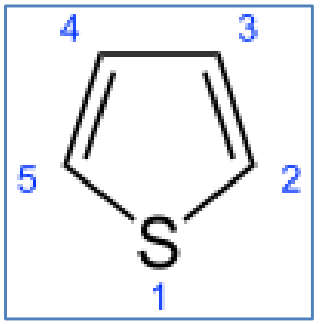
Pyridine



Pyrazine



Pyrazole



Thiophene

Note: The aromatic compounds which do not contain benzene ring in their structures are called as a **non-benzenoid aromatics or non-benzenoid aromatic compound.**

- In all these examples, the number of π-electrons is 6, due to the π electrons from the double bonds as well as the two electrons from any lone pair that is in the p-orbital that is in the plane of the aromatic π system.

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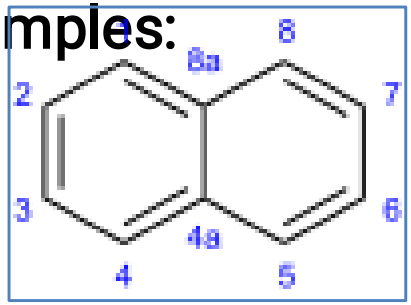
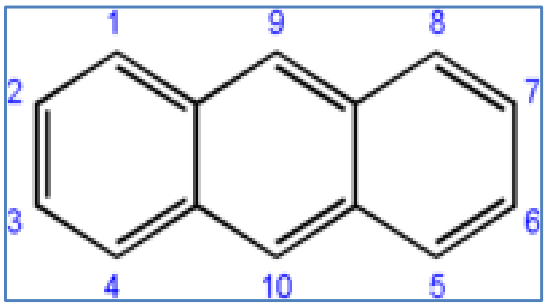
[3]. Fused aromatics and polycyclics:

- Polycyclic aromatic hydrocarbons are molecules containing two or more simple aromatic rings fused together by sharing two neighboring carbon atoms.

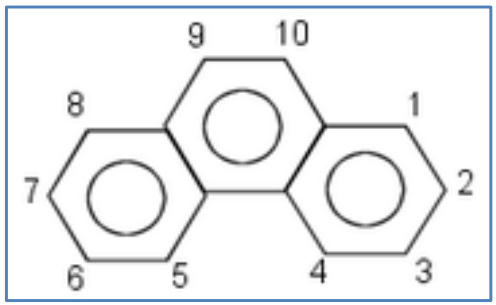
Examples:



Naphthalene



Anthracene

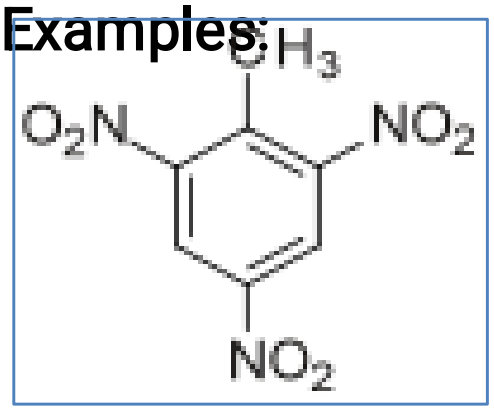


Phenanthrene

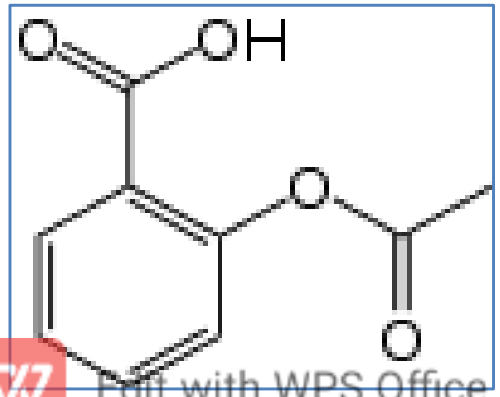
[4]. Substituted Aromatics:

- Many chemical compounds are aromatic rings with other functional groups attached.

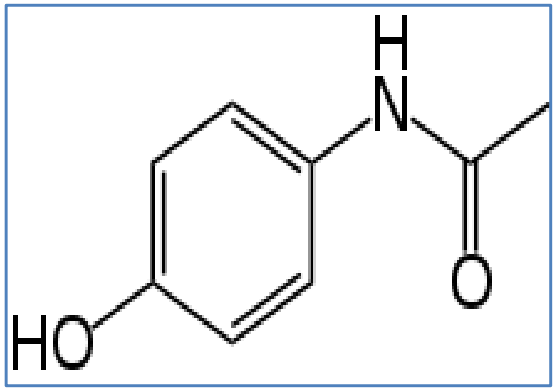
Examples:



Trinitrotoluene (TNT)



Acetylsalicylic Acid (Aspirin)



Paracetamol

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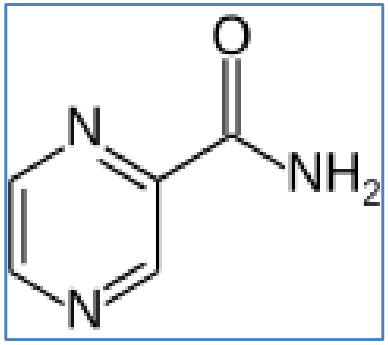
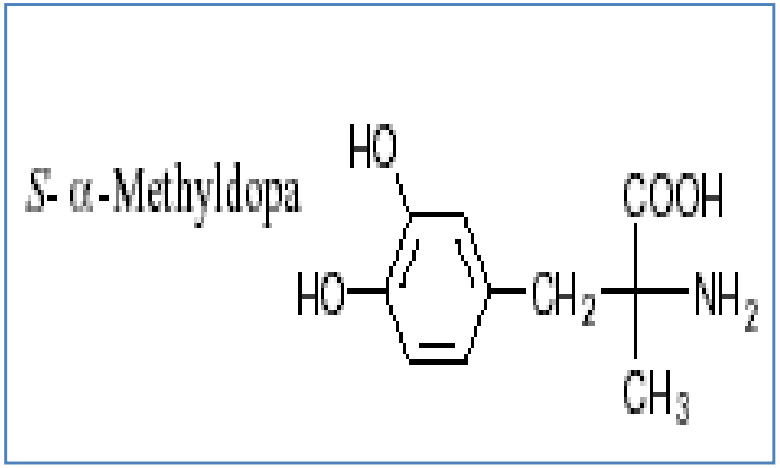
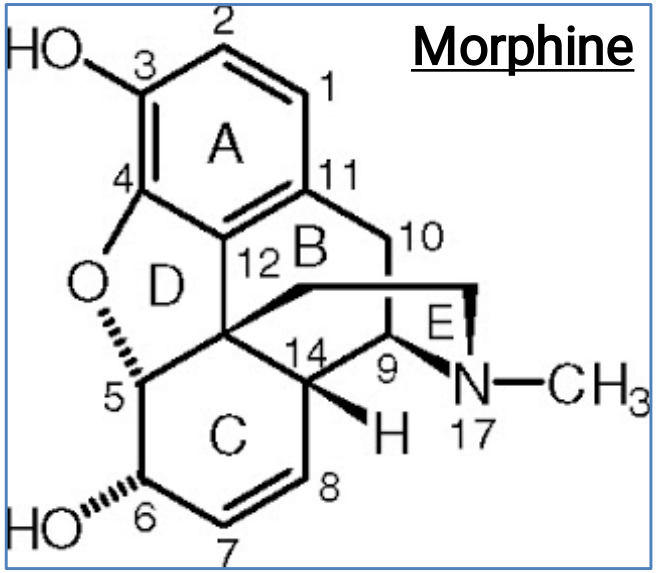
Importance of Aromatic Compounds

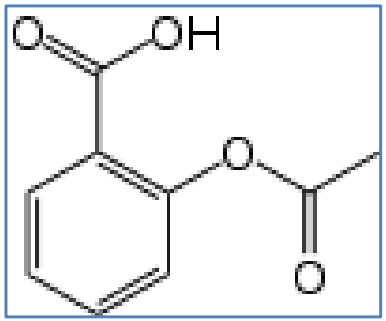
[1]. Many drugs like **aspirin** (NSAID), **S-α-Methyldopa** (anti-hypertensive drug), **pyrazinamide** (anti TB), **morphine** (opioid analgesic), **paracetamol** (NSAID) etc. have aromatic structures.



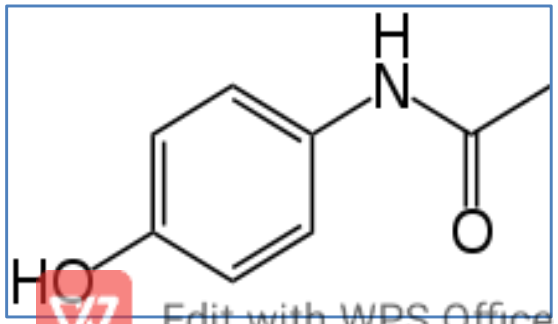
Pyrazinamide







Acetylsalicylic Acid (Aspirin)



Paracetamol

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2. Aromatic molecules typically display **enhanced chemical stability,** compared to similar non-aromatic molecules. This extra stability has a profound effect on the chemistry of an aromatic molecule.

3. Aromatic compounds **undergo** electrophilic aromatic substitution and nucleophilic aromatic **substitution reactions, but not** electrophilic **addition reactions** as happens with simple carbon–carbon double bonds.

4. Aromatic compounds play **key roles in the biochemistry** of all living things. **For example:**
   - The four **aromatic amino acids** histidine, phenylalanine, tryptophan, and tyrosine each serve as one of the 20 basic building-blocks of proteins.
   - Further, all 5 nucleotides (adenine, thymine, cytosine, guanine, and uracil) that make up the sequence of the genetic code in DNA and RNA are aromatic purines or pyrimidines.

   - The molecule **heme** contains an aromatic system with 22 π-electrons.

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Topic:

Tautomerism of Carbonyl Compounds

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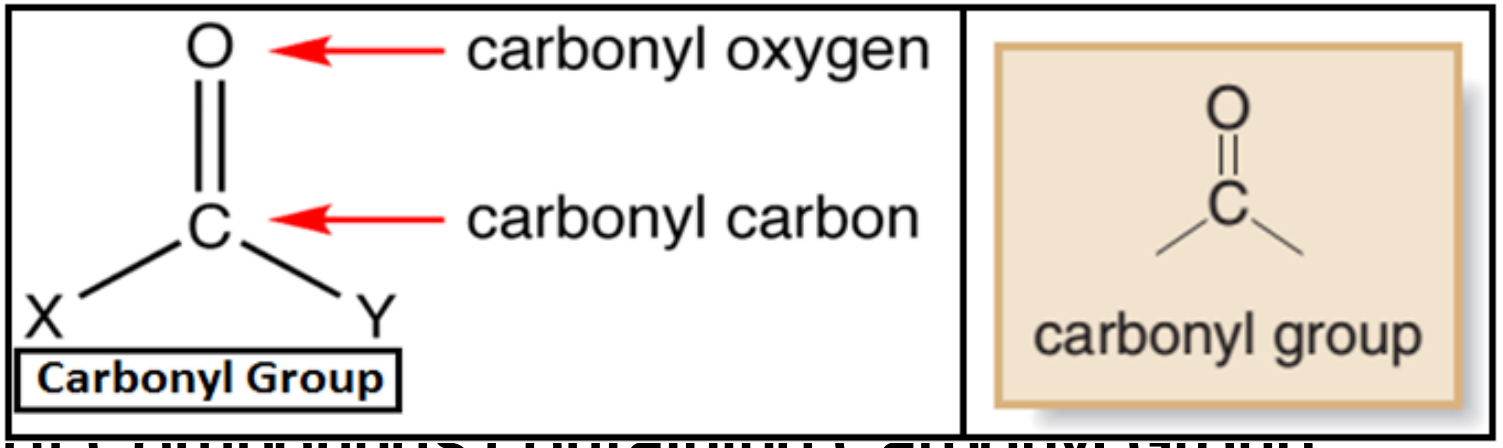
Introduction:

- **Carbonyl Compound:** "A compound containing a **carbonyl group** is often referred to as a carbonyl compound."

- **Carbonyl group:** "It is a functional group composed of a carbon atom double bonded to an oxygen atom."



Classes of compounds containing carbonyl group:

- Two broad classes of compounds contain a carbonyl group:

1. Compounds that have only carbon and hydrogen atoms bonded to the carbonyl group **i.e. aldehydes & ketones**
2. Compounds that contain an electronegative atom bonded to the carbonyl group. These include **carboxylic acids, acid chlorides, esters and amides,** as well as other similar compounds.

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Examples:

1. Carbonyl compounds in which carbonyl group is bonded to a carbon and at least one H atom are known as **aldehydes.** In aldehydes, is on the "end" of a carbon chain.

2. Carbonyl compounds in which carbonyl group is bonded to two alkyl or aryl groups are known as **ketones.** In ketones, the carbonyl group is in the "middle" of a carbon chain.

3. The carbonyl compounds in which carbonyl group is bonded to oxygen are known as **carboxylic acids, and their derivatives (e.g. esters, anhydrides).**

4. Carbonyl compounds where carbon is attached to nitrogen are called **amides.**

5. Carbonyl compounds where carbon is attached to halogen are called **acyl halides.**

Importance:

- The carbonyl group is perhaps the **most important functional group in organic chemistry** because its electron def ̄cient carbon and easily breakable π bond make it susceptible to a wide variety of useful reactions.

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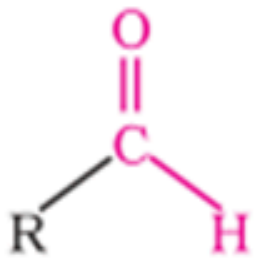
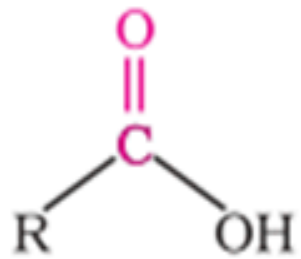
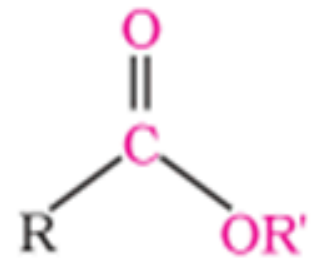
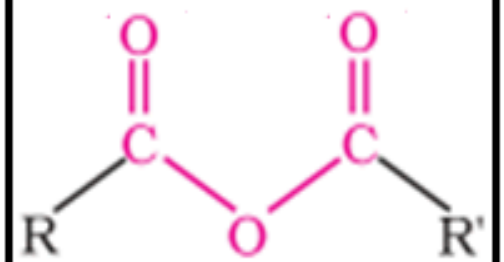
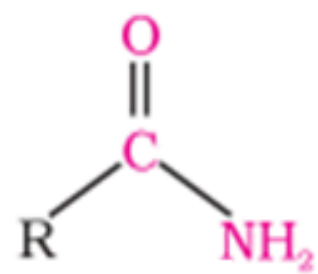
Some of the types of compounds characterized by carbonyl group :

General Formula				
Class	Aldehyde	Ketone	Carboxylic acid	Ester
General Formula				
Class	Acyl halide; X = (Halogen)	Acid anhydride	Amide	

- **Enol Group:** An alcoholic group on C=C is called **enol.** It is an **alkene alcohol**.

- **Important Note:** A carbonyl compound with a **hydrogen atom on its α carbon** rapidly equilibrates with its corresponding **enol** isomer.

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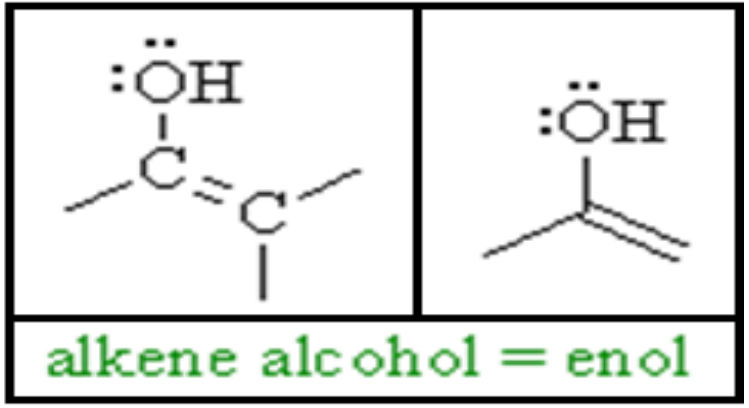
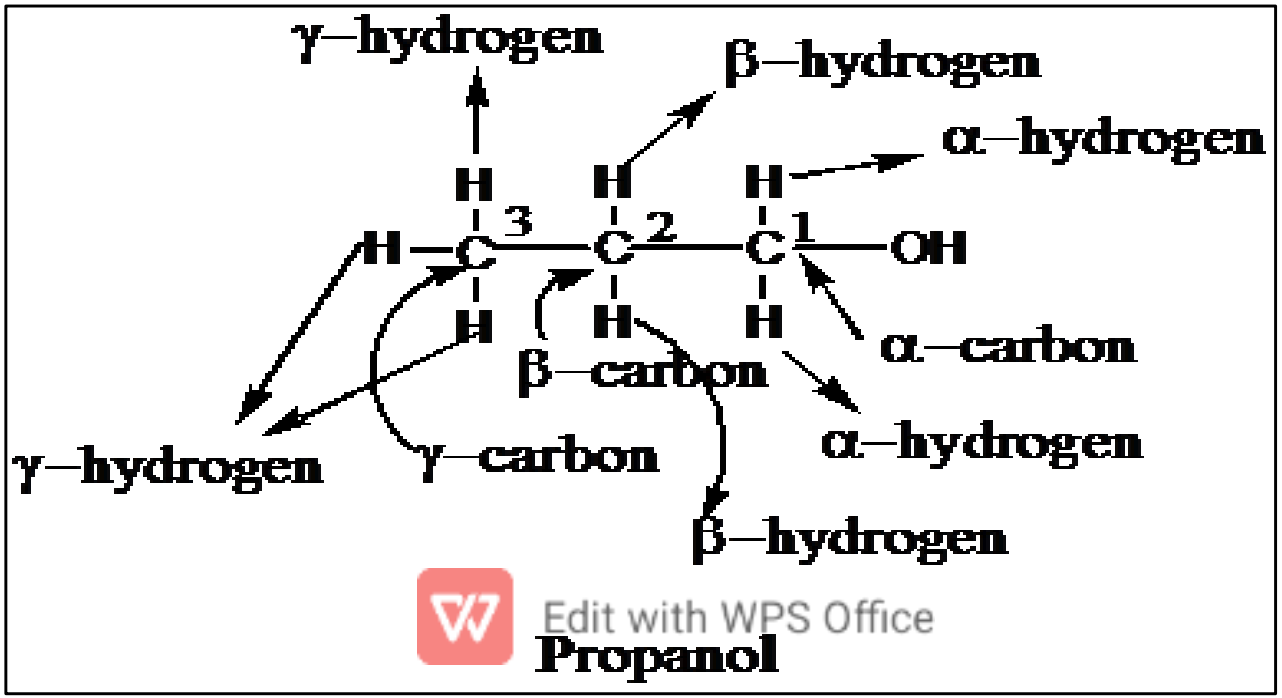


- **Alpha Carbon:** The alpha carbon (C_α) in organic molecules refers to the first carbon atom that attaches to a functional group, such as a carbonyl.

- The second carbon atom is called the beta carbon (C_β), and the system continues naming in alphabetical order with Greek letters.



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Tautomerism: "Two molecules with the same molecular formula but different connectivity (functional groups) which can **interconvert in a rapid equilibrium** are called **tautomers** and the phenomena of their interconversion is called **tautomerism.**"

- "Tautomers **are constitutional isomers** of organic compounds that readily interconvert with each other.

- In many cases tautomerism occurs in compounds merely by **exchanging a hydrogen atom** between two other atoms, to either of which it forms a covalent bond.

- In simple words, tautomers are constitutional isomers that differ by the location of a proton and a pi bond.

Most Common Type of Tautomerism:

- The most common type of tautomerism is **keto-enol tautomerism** which involves keto compounds and unsaturated hydroxyl compounds (enols).

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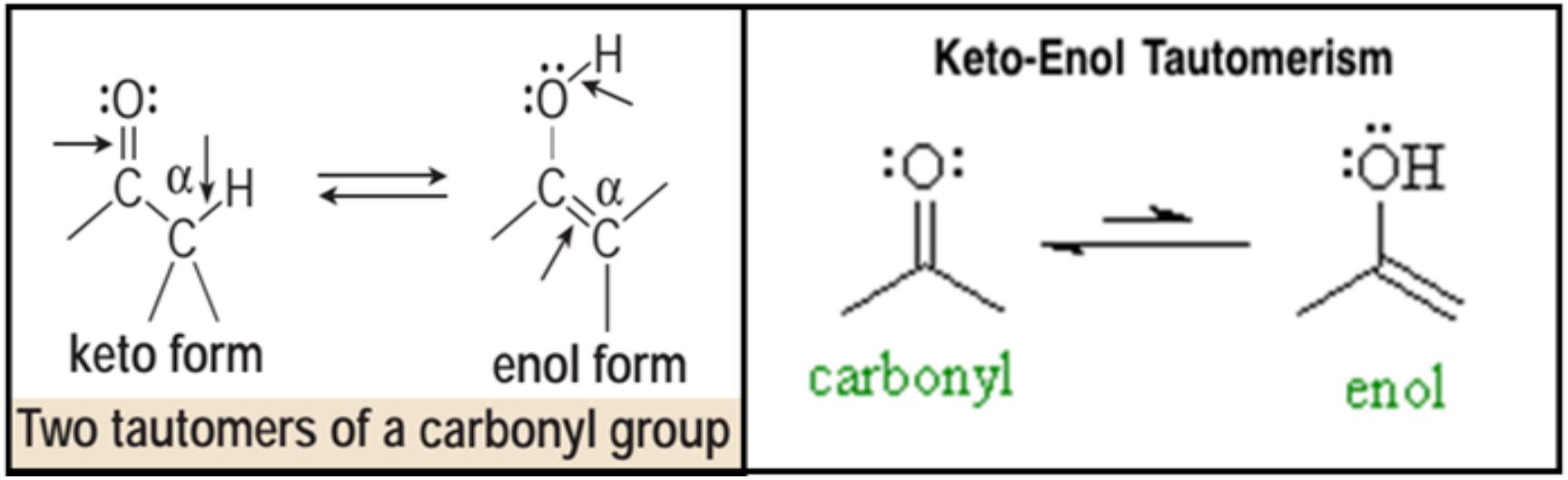
- **Keto-enol Tautomerism:** "Keto-enol tautomerism refers to a chemical equilibrium between a keto form (a ketone or an aldehyde) and an enol form (an alcohol) of a compound." It is most often **a special case of functional group isomerism**.

- Keto-enol tautomerism can occur with carbonyl compounds if the carbon adjacent to the carbonyl, the α carbon, possesses a hydrogen. **Such α hydrogens are acidic.**

- The carbonyl group can be converted to enol form due to transfer of one of the **α-hydrogen** onto the carbonyl oxygen atom.



- **Enol and keto forms** are tautomers of the carbonyl group that differ in the position of the double bond and a proton.

- These are constitutional isomers which remain in equilibrium with each other and are easily interconverted by a trace of acid or base.

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Keto and enol Tautomers:

- The enol and keto forms are said to be tautomers of each other.
- A **keto tautomer** has a C=O and an additional C–H bond.
- An **enol tautomer** has an O–H group bonded to a C=C. The structural change is the shift of a hydrogen atom between atoms of carbon and oxygen, with the rearrangement of bonds.

More Stable Tautomer:

- Equilibrium favors the **keto form for most carbonyl compounds** largely because a C=O is much stronger than a C–C.
- For simple carbonyl compounds, < 1% of the enol is present at equilibrium. With unsymmetrical ketones, moreover, two different enols are possible, yet they still total < 1%.

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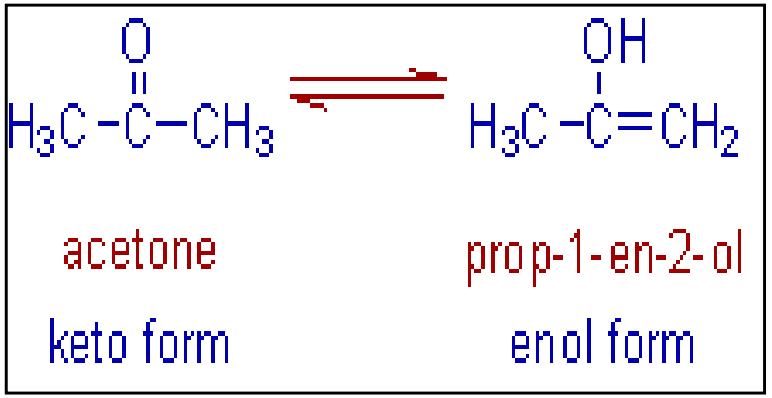
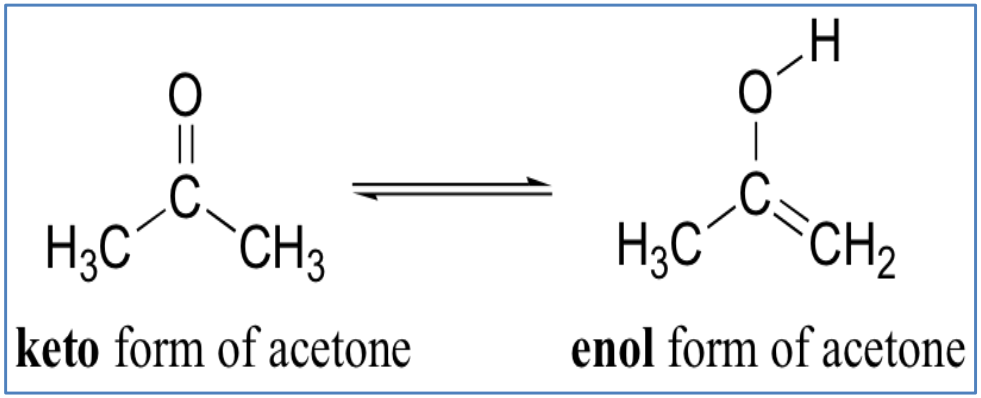
- **Example of a Ketone Undergoing Keto-enol Tautomerism:** The dynamic equilibrium between a keto form and an enol form exhibited by **acetone** is shown below.



OR



- However the relatively stable keto form exists in higher percentage (about 99%).

- The % composition of keto enol tautomeric mixture depends on the relative stabilities of these two forms.

- **Example of an Aldehyde Undergoing Keto-enol Tautomerism :** If an aldehyde possesses at least one hydrogen atom on the carbon atom adjacent to the carbonyl group, called the alpha (α) carbon, this hydrogen can migrate to the oxygen atom of the carbonyl group.

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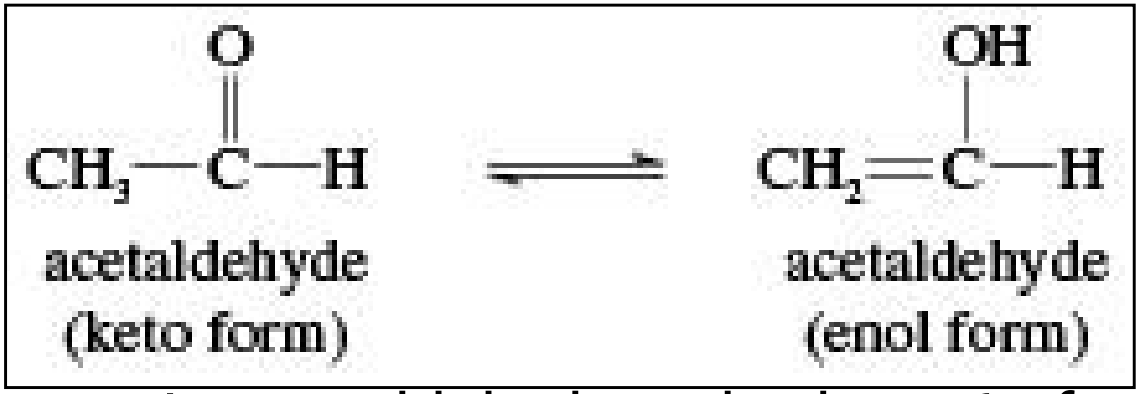
- The double bond then migrates to the α-carbon. As a result, a carbonyl compound with an α-hydrogen can exist in two isomeric forms, called tautomers.

- In the keto form, the hydrogen is bonded to the α-carbon, while in the enol form it is bonded to the carbonyl oxygen with the migration of the double bond.



- **Abundant Tautomer:** In acetaldehyde, only about 6 of every 10 million molecules are in the enol form at any given time.

- Nevertheless, the equilibrium always exists, and every molecule of acetaldehyde is converted to the enol form (and back again) several times per second.

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The Mechanism of Tautomerization or Tautomerism:

- Tautomerization, the process of converting one tautomer into another, is catalyzed by both acids and bases.

- Tautomerization always requires two steps (protonation and deprotonation), but the order of these steps depends on whether the reaction takes place in acid or base.

- In the mechanisms discussed below for tautomerization, the keto form is converted to the enol form.

- All of the steps are reversible, though, so they equally apply to the conversion of the enol form to the keto form.

[1]. Acid-catalyzed Tautomerization:

- With acid, protonation precedes deprotonation.

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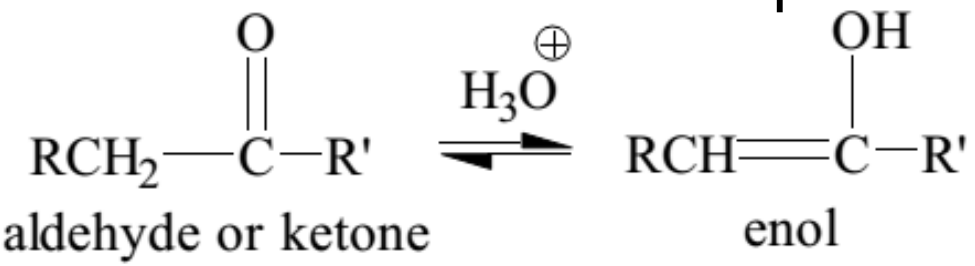
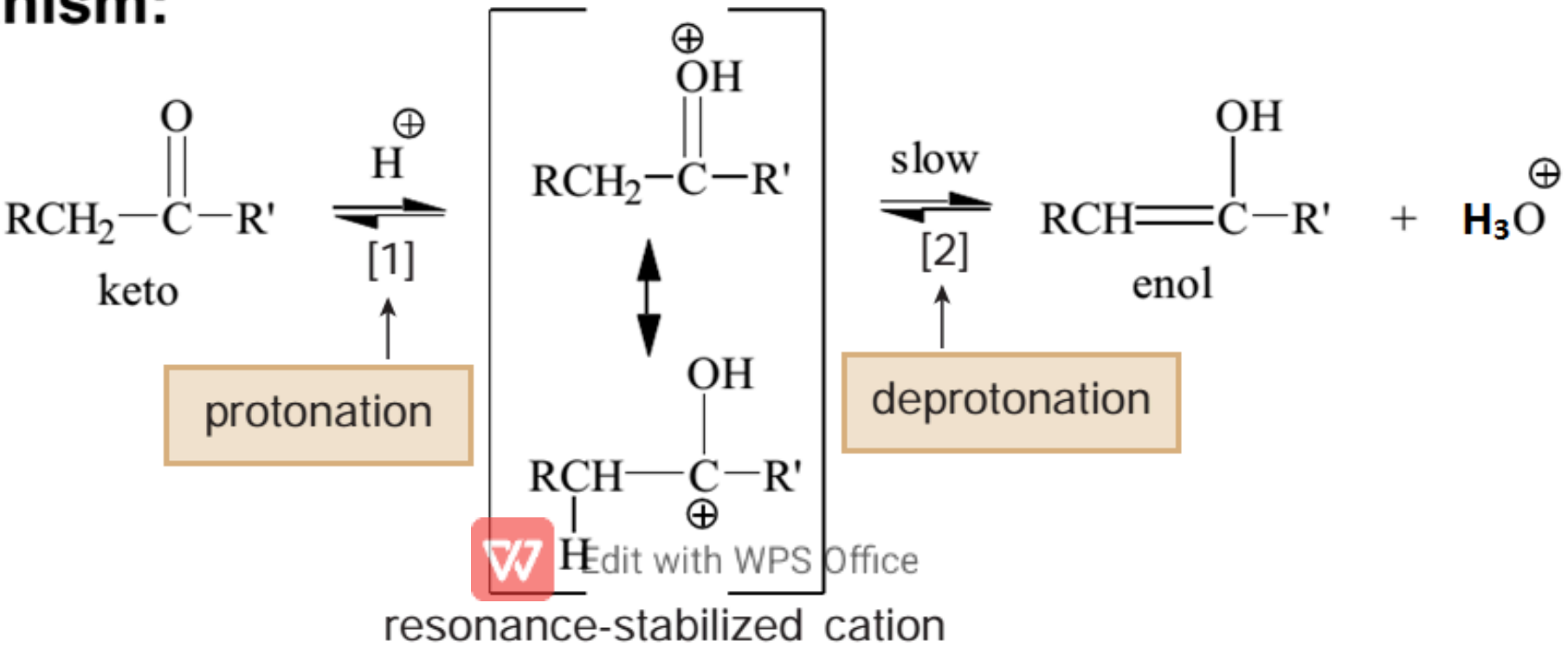
- **Protonation of the carbonyl oxygen** forms a resonance-stabilized cation in Step [1], and **deprotonation in Step [2]** forms the enol. The net result of these two steps is the movement of a double bond and a proton.

- In Acid-catalyzed enolization of aldehydes or ketones the removal of the α-hydrogen is still slow but occurs in the 2nd step.

Overall Reaction:



Mechanism:



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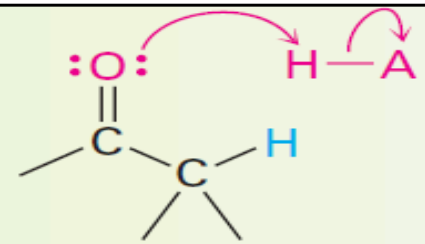
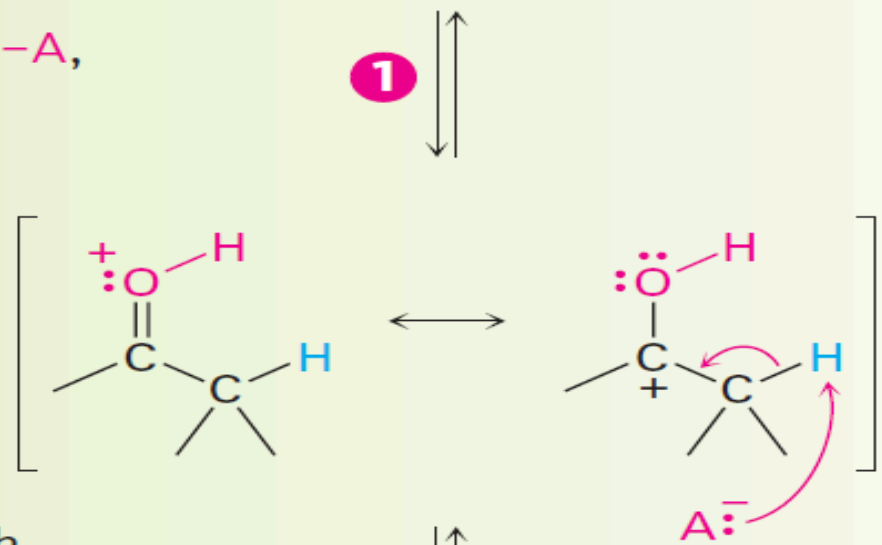
Alternate Explanation

(a) Acidic conditions

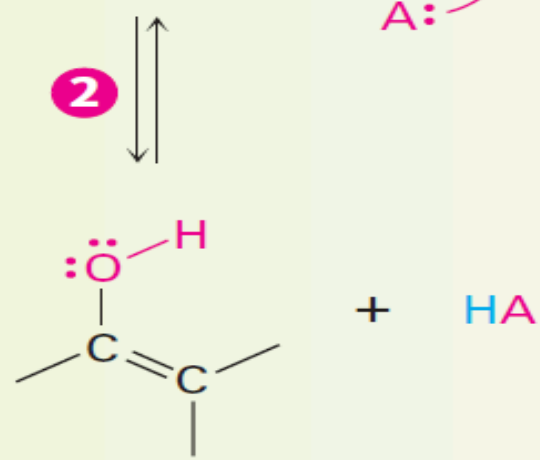


Keto tautomer

1. The carbonyl oxygen is protonated by an acid H–A, giving a cation with two resonance structures.



2. Loss of H^+ from the α position by reaction with a base A^- gives the enol tautomer and regenerates HA catalyst.



Enol tautomer

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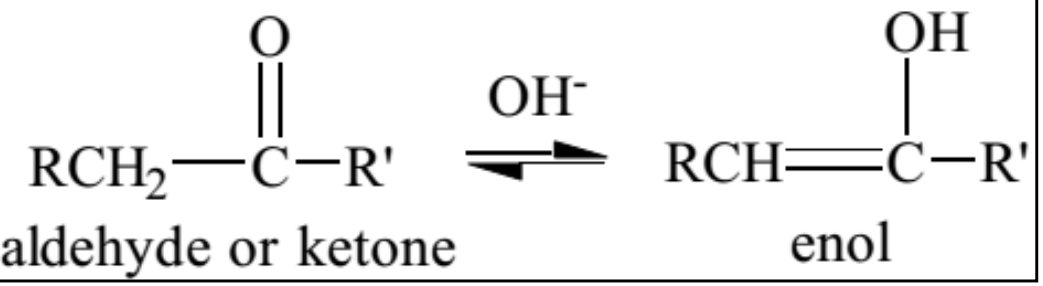
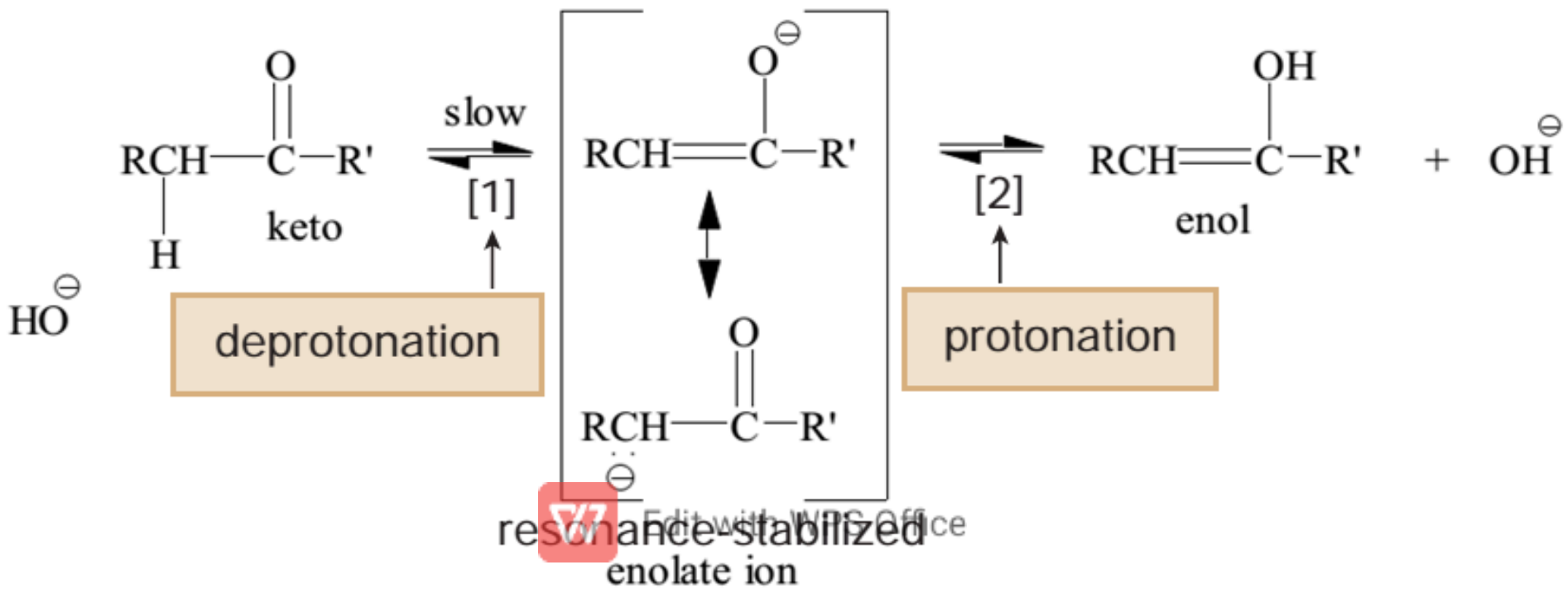
[2]. Tautomerization in Base: With base, deprotonation precedes (occurs before) protonation. Carbonyl compound can act as an acid and donate one of it's α-hydrogens to a suf ficiently strong base, yielding an enolate ion in Step [1] which is resonance-stabilized. Protonation of the enolate ion with H_2O forms the enol in Step [2].

Overall Reaction:



Mechanism:



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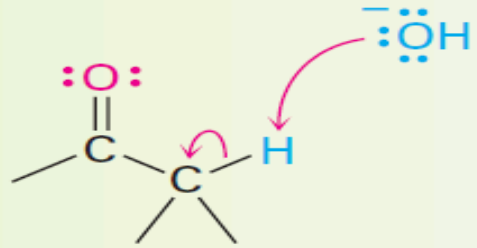
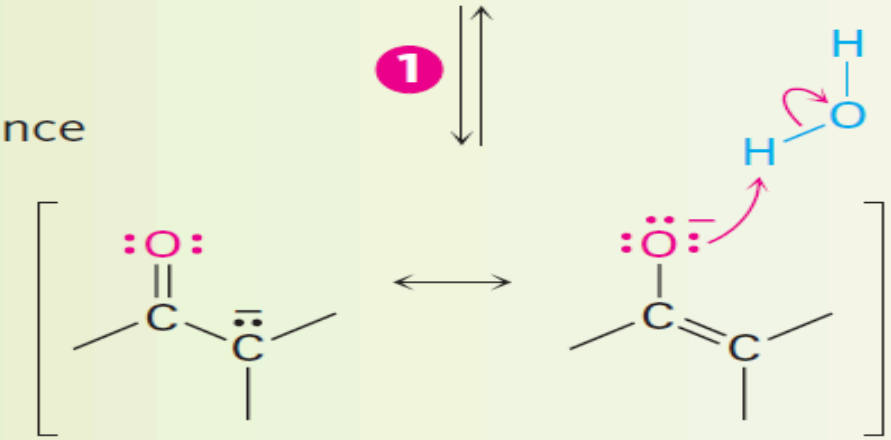
Alternate Explanation

(b) Basic conditions



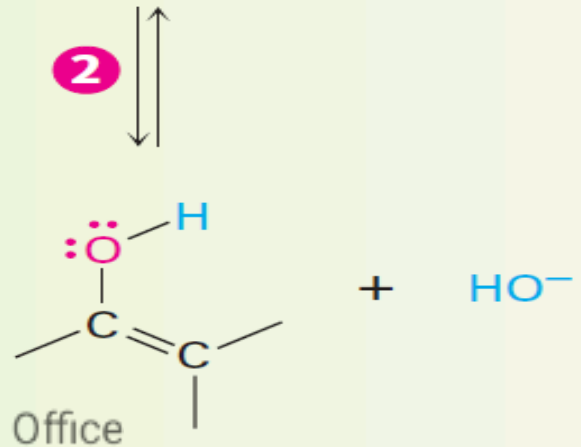
Keto tautomer

1. Base removes the acidic α hydrogen, yielding an enolate ion with two resonance structures.



Enolate ion

2. Protonation of the enolate ion on oxygen gives the enol and regenerates base catalyst.



Enol tautomer

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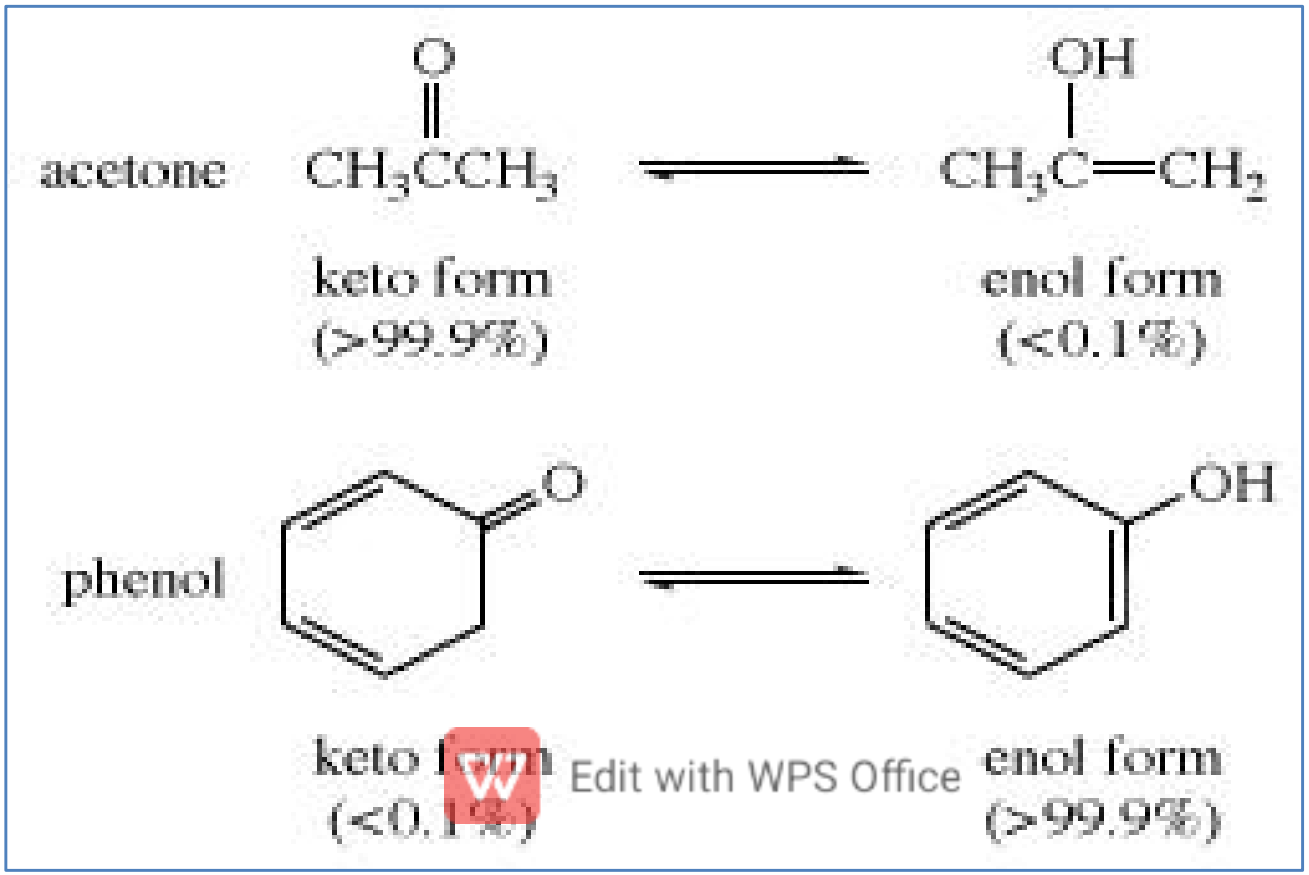
Predominant Tautomer

- **In general** the **keto form is the low energy form and is more stable** than the enol form e.g. in many aliphatic aldehydes and ketones, such as acetone, the keto form is the predominant one at equilibrium. Enol tautomer is often present to a small extent and cannot be isolated easily.

- However the **enol form is also stable in certain cases due to other stability** factors e.g. in phenols, the enol form, which is stabilized by the aromatic character of the benzene ring, is the major component.



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Factors Shifting the Equilibrium toward Enol Form

- Although the *keto* form is most stable for aldehydes and ketones in most situations, there are several factors that will shift the equilibrium toward the *enol* form.

- The same factors that stabilize alkenes or alcohols will also stabilize the enol form.

- There are two strong factors and three subtle factors.

Two strong factors:

1. **Aromaticity**. Phenols can theoretically exist in their keto forms, but the enol form is greatly favored due to aromatic stabilization.

2. **Hydrogen Bonding.** Nearby hydrogen bond acceptors stabilize the enol form. When a Lewis basic group (which can donate pair(s) of electrons) is nearby, the enol form is stabilized by internal hydrogen bonding.

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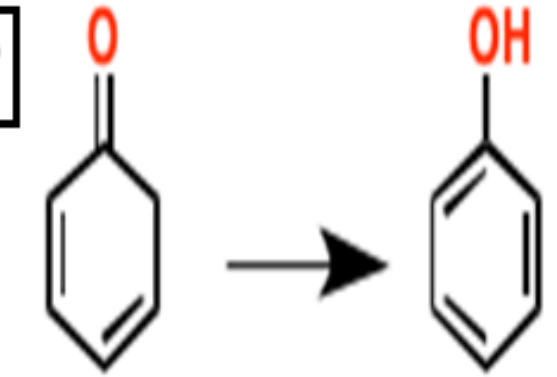
Case 1: Aromaticity

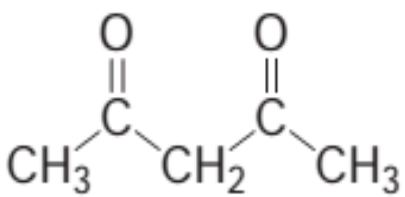


Phenone → **Phenol**

Not observed | *Exclusive*

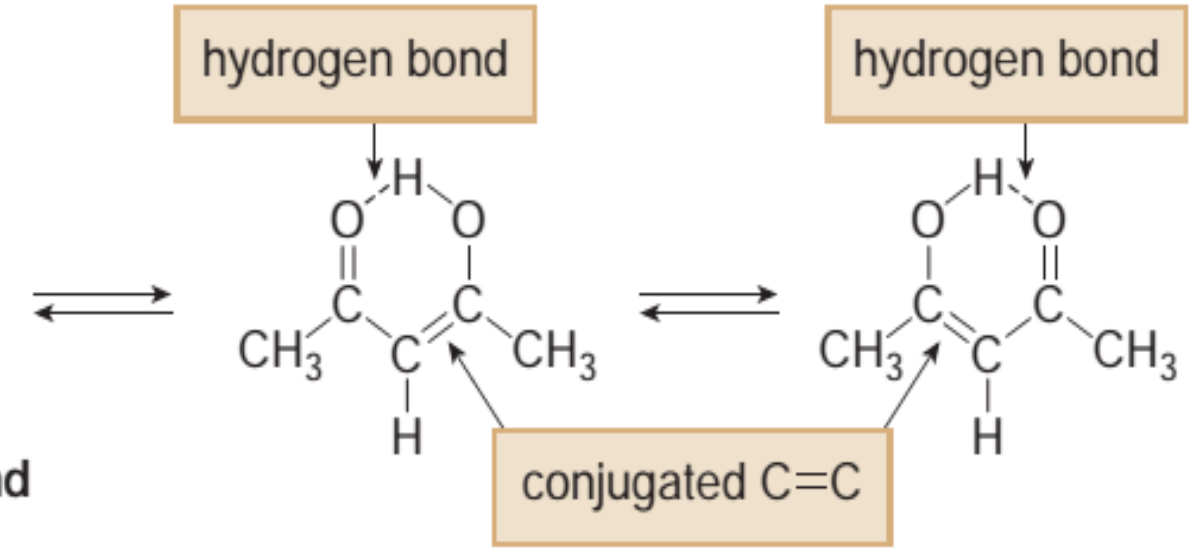
Ketones will go completely to the enol form if they are stabilized by aromaticity.

Case 2: H-Bonding



2,4-pentanedione
β-dicarbonyl compound

24% keto tautomer



76% enol tautomers

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Three subtle (indirect) effects:

[1]. Solvent:

- Solvent can also play an important role in the relative stability of the enol form.

- **For example,** in benzene, the **enol form of 2,4-pentanedione** predominates in a 94:6 ratio over the keto form, whereas the numbers almost reverse completely in water.

- **Reason:** In a polar protic solvent like water, the lone pairs will be involved in hydrogen bonding with the solvent, making them less available to hydrogen bond within the enol form.

[2]. Conjugation:

- π systems want to connect together than hang out in isolation.

- So in the example given below, the more favorable tautomer will be the one on the left, where the double bond is connected by conjugation to the phenyl.

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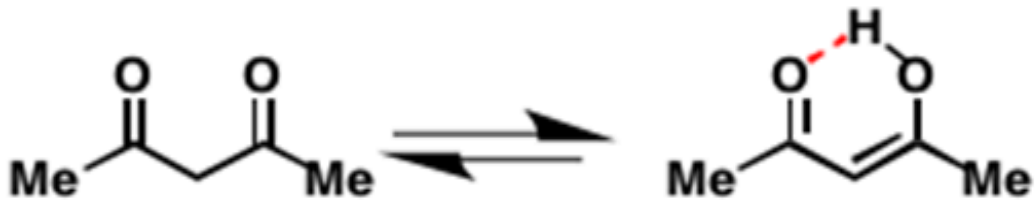
Tautomerism: More subtle effects

Case 3: Solvent



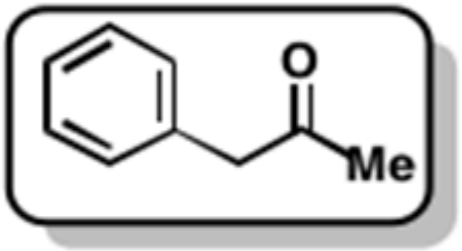
Hydrogen bonding solvent will make lone pairs less available for bonding.

6 : 94 (using benzene as solvent)

81 : 19 (using water as solvent)

Case 4: Conjugation

Example: Which enol form is favored for ketone



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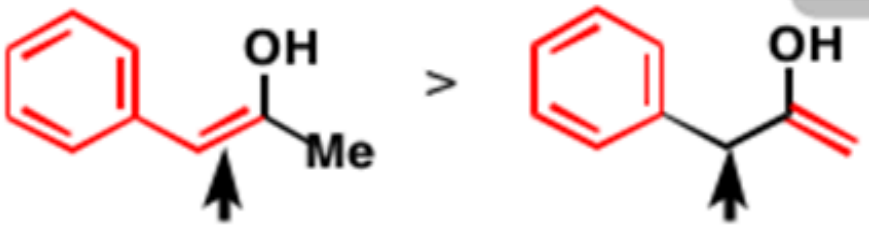

double bond is in conjugation with phenyl

double bond **not** in conjugation with phenyl

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For your concept: Lewis Definitions of Acids and Bases

- A **Lewis acid** is any substance (e.g. H^+ ion) that can accept a pair of nonbonding electrons. In other words, a **Lewis acid** is an **electron-pair acceptor.**

- A **Lewis base** is any substance (e.g. OH^- ion) that can donate a pair of nonbonding electrons. A Lewis base is therefore an **electron-pair donor.**

$$Al^{3+}(aq) + 6\,H_2O(l) \rightleftharpoons Al(H_2O)_6^{3+}(aq)$$

- This is an example of a Lewis acid-base reaction.

- The Lewis structure of water suggests that this molecule has nonbonding pairs of valence electrons and can therefore act as a Lewis base when an Al^{3+} ion acting as a Lewis acid picks up six pairs of electrons from neighboring water molecules acting as Lewis bases to give an **acid-base complex**, or **complex ion**.

Protic solvent:

- A **protic solvent** is a solvent that has a hydrogen atom bound to an oxygen (as in a hydroxyl group) or a nitrogen (as in an amine group).

- In general terms, any solvent that contains labile H is called a protic solvent. The molecules of such solvents readily donate protons (H^+) to reagents.

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Thank you Very Much

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